Converting a CFG to an NPDA

Theorem: For every context-free language L there exists a nondeterministic PDA, M, such that M exactly recognizes L, and conversely.

Algorithm: CFG to an NPDA

The machine M will be nondeterministic, will have only one state, q, and will have the transition function rules defined as follows:

- $\delta(q, \lambda, A)$ contains (q, w) for every production $A \rightarrow w$ in P.
- $\delta(q, a, a)$ contains (q, λ) for every $a \in \Sigma$.

The stack symbols are in $V \cup \Sigma$ and the stack initially contains the start symbol S. Machine M will halt by empty stack.

Example: Consider the following context-free grammar with $\Sigma = \{a, b, c\}$.

$S \rightarrow aSb \mid c$

The transition function rules are:

$\delta(q, a, a) = \{(q, \lambda)\}$

$\delta(q, b, b) = \{(q, \lambda)\}$

$\delta(q, c, c) = \{(q, \lambda)\}$

$\delta(q, \lambda, S) = \{(q, aSb), (q, c)\}$